

# ADSA Analysis Report

Task: Classification — Target: Churn

Best Model: XGBoost

## Executive Summary

- Task identified as classification for predicting "Churn".
- XGBoost achieved 77.4% accuracy and 74.8% F1-score on the holdout set.
- The most influential factors are: ContractType\_One Year, ContractType\_Two Year, MonthlyCharge.
- Strong correlation (0.84) found between TenureMonths and TotalCharges.

## Data Profile

| Metric         | Value |
|----------------|-------|
| Rows           | 5012  |
| Columns        | 18    |
| Duplicate Rows | 12    |
| Memory (MB)    | 3.14  |

## Data Cleaning Summary

- convert\_datetime: JoinDate ()
- drop\_duplicates: ()
- impute: Age (mean)
- impute: MonthlyCharge (median)
- impute: DataUsageGB (median)
- clip\_outliers: MonthlyCharge (15)
- clip\_outliers: TotalCharges (44)
- clip\_outliers: DataUsageGB (65)

## Model Evaluation

| Metric    | Value  |
|-----------|--------|
| Accuracy  | 0.7740 |
| Precision | 0.7370 |
| Recall    | 0.7740 |
| F1_Score  | 0.7476 |

## Cross-Validation Results

| Model              | CV Score |
|--------------------|----------|
| LogisticRegression | 0.5392   |
| RandomForest       | 0.5235   |
| XGBoost            | 0.5602   |

# Visualizations







